Page No.
Date

Day: 08

* What is an IP Address?

- An IP address is a 32-bit number that uniquely identifies a device on a network. Written in dotted decimal notation — 4 octets separated by dots.

192 . 168 . 1 . 1
11000000.10101000.00000001.00000001

* Two Parts of Every IP Address

- Network Portion: Identifies the network
- Host Portion: Identifies the device on that network

The subnet mask tells you where the line is drawn.

* Subnet Mask

- A subnet mask is also a 32-bit number. It uses 1s for the network part and 0s for the host part.

CIDR	/8	/16	/24	/32
Subnet Mask	255.0.0.0	255.255.0.0	255.255.255.0	255.255.255.255
Network bits	8	16	24	32
Host bits	24	16	8	0

Page No.
Date

* CIDR Notation

- CIDR (Classless Inter-Domain Routing) = IP address + prefix length

192.168.1.0/24
↑ 24 bits are the network portion.

* Key Formulas

1) Total Hosts = $2^{(\text{host bits})}$

2) Usable Hosts = $2^{(\text{host bits})} - 2$ (N/w & broadcast)

3) Network Address = First IP in the range (all host bits = 0)

4) Broadcast Address = Last IP in the range (all host bits = 1)

5) First Usable Host = Network Address + 1

6) Last Usable Host = Broadcast Address − 1

* Step-by-Step Subnetting: 1) 192.168.1.0/24

1) Host bits = 32 − 24 = 8 bits

2) Total Hosts = 2^8 = 256

3) Usable Hosts = 256 − 2 = 254

4) Network Address = 192.168.1.0

5) Broadcast Address = 192.168.1.255

6) First Usable = 192.168.1.1

7) Last Usable = 192.168.1.254

8) Subnet Mask = 255.255.255.0

2) 10.0.0.0/8

1) Host bits = 32 − 8 = 24

2) Total hosts = 2^{24} = 16777216

3) Usable hosts = 16777216 − 2 = 16777214

4) Network Address = 10.0.0.0

5) Broadcast Address = 10.255.255.255

6) First Usable = 10.0.0.1

7) Last Usable = 10.255.255.254

8) Subnet Mask = 255.0.0.0

3) 172.16.5.43/20

1) Host Bits = 32 − 20 = 12

2) Total Hosts = 2^{12} = 4096

3) Usable Hosts = 4096 − 2 = 4094

4) Network Address = 172.16.0.0

5) Broadcast Address = 172.16.15.255

6) First Usable = 172.16.0.1

7) Last Usable = 172.16.15.254

8) Subnet Mask = 11111111.11111111.11110000.00000000
(255.255.240.0)

* Private IP Address Ranges

Range	CIDR	Used for
10.0.0.0 – 10.255.255.255	/8	Large Networks
172.16.0.0 – 172.31.255.255	/12	Medium Networks
192.168.0.0 – 192.168.255.255	/16	Home / small Networks

Note: Your home WiFi is 192.168.x.x – that's why you see it in Wireshark!

* Why Subnetting Matters for DFIR

- Every IP in a PCAP belongs to a network – subnetting tells you which network.

Page No.
Date

- 192.168.x.x = private / internal traffic.
- 8.8.8.8 , 142.250.x.x = public/ external traffic.
- knowing subnets helps you identify lateral movement — attacker moving between internal subnets.
- RFC 1918 private ranges should never appear as destination IPs on the internet — if they do, something is wrong.

SubnettingPractice.com Free Cheatsheet Questions Calculators H

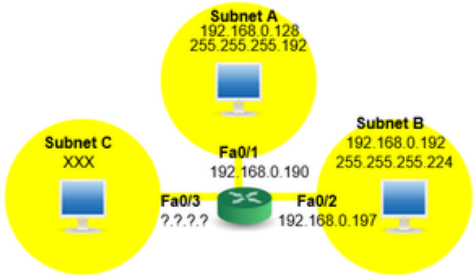
Subnetting Questions

Using the diagram below, which of the following IP addresses would be valid to asign to interface Fa0/3? For your answer type out the IP address.

192.168.0.114

192.168.0.131

192.168.0.201



192.168.0.114 Submit

Show Answer Next

☐ I want only this type of question.

Yes! The correct answer is: